Lab 5

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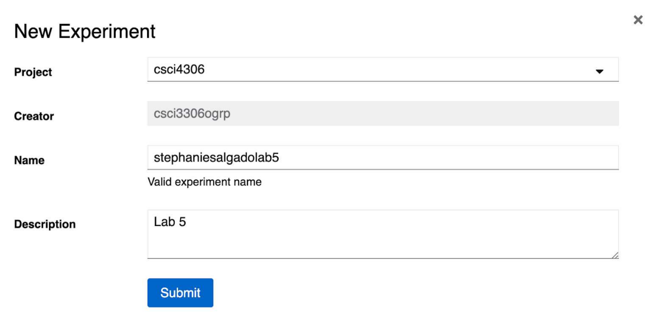
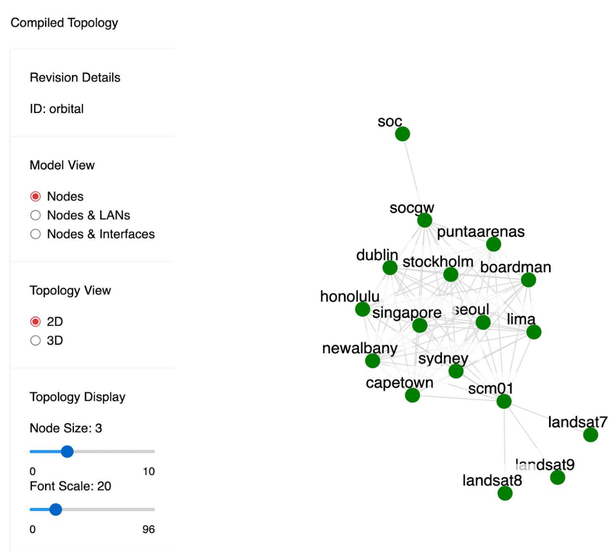
Setup:

Creating an Experiment



Like in previous labs, I created a new experiment from the “Experiments” tab.

Compiled Topology

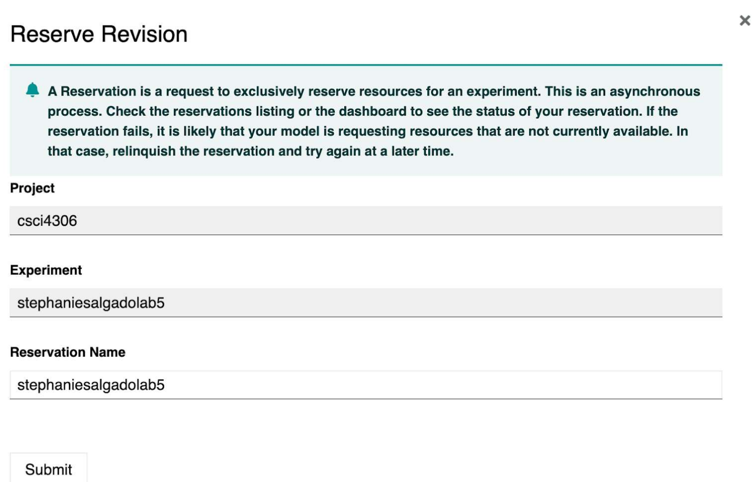


I use the provided model and I compile it in the “Model Editor” tab.

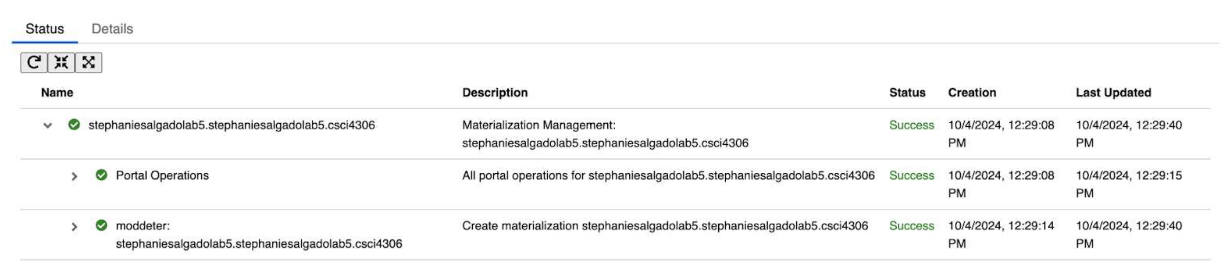
Pushing Model to Experiment

Then, I push the model to the new experiment for Lab 5.

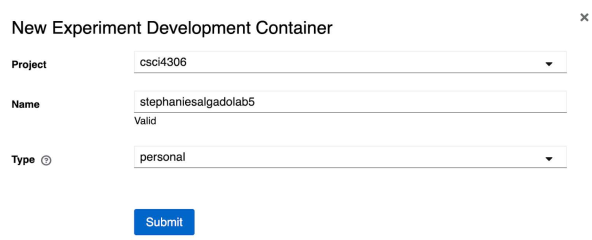
Realizing the Experiment

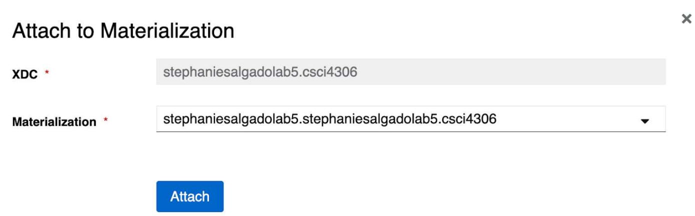


Materializing the Realization



Attaching XDC





After all these steps, I successfully completed the setup for this lab.

Accessing all nodes

```
# su - csci3306ogrp
csci3306ogrp@stephaniesalgadolab5:~$ ssh capetown
Warning: Permanently added 'capetown' (ED25519) to the list of known ho
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-182-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

csci3306ogrp@capetown:~$
```

To access the nodes, I use “su - csci3306ogrp” then I use “ssh” followed by the name of the node I want to log into. I do the same for all nodes.

Installing “net-tools” on all nodes

```
csci3306ogrp@boardman:~$ sudo apt install net-tools
sudo: unable to resolve host boardman.infra.stephaniesalgadolab5.stephaniesalgadolab5.csci430: Name or service not known
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  net-tools
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 196 kB of archives.
After this operation, 864 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu focal/main amd64 net-tools amd64 1.60+git20180626.aebd88e-1ubuntu1 [196 kB]
Fetched 196 kB in 1s (241 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 105037 files and directories currently installed.)
Preparing to unpack .../net-tools_1.60+git20180626.aebd88e-1ubuntu1_amd64.deb ...
Unpacking net-tools (1.60+git20180626.aebd88e-1ubuntu1) ...
Setting up net-tools (1.60+git20180626.aebd88e-1ubuntu1) ...
Processing triggers for man-db (2.9.1-1) ...
csci3306ogrp@boardman:~$
```

Then, I use “sudo apt install net-tools” on all nodes.

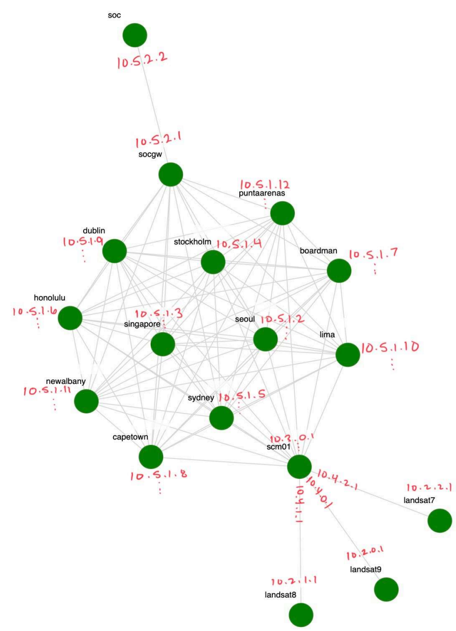
Using “route” and “ifconfig” for all nodes

```
csci3306ogrp@seoul:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         172.30.0.1      0.0.0.0         UG    512    0        0 eth0
10.1.0.0        0.0.0.0         255.255.255.0   U     0      0        0 eth1
10.5.1.0        0.0.0.0         255.255.255.0   U     0      0        0 eth2
172.30.0.0      0.0.0.0         255.255.0.0     U     0      0        0 eth0
172.30.0.1      0.0.0.0         255.255.255.255 UH    512    0        0 eth0
192.168.254.0   172.30.0.1      255.255.255.0   UG    99     0        0 eth0
csci3306ogrp@seoul:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.30.0.22  netmask 255.255.0.0  broadcast 172.30.255.255
        inet6 fe80::5468:53ff:fe9c:bf7f  prefixlen 64  scopeid 0x20<link>
        ether 56:68:53:9c:bf:7f  txqueuelen 1000  (Ethernet)
        RX packets 1010  bytes 536300 (536.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 405  bytes 47347 (47.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.1.0.1  netmask 255.255.255.0  broadcast 10.1.0.255
        inet6 fe80::cd8:34ff:fe46:213a  prefixlen 64  scopeid 0x20<link>
        ether 0e:d8:34:46:21:3a  txqueuelen 1000  (Ethernet)
        RX packets 37  bytes 2806 (2.8 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 15  bytes 1146 (1.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.5.1.2  netmask 255.255.255.0  broadcast 10.5.1.255
        inet6 fe80::fc45:8bff:fe86:f60d  prefixlen 64  scopeid 0x20<link>
        ether fe:45:8b:86:f6:0d  txqueuelen 1000  (Ethernet)
        RX packets 184  bytes 14072 (14.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 15  bytes 1146 (1.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
csci3306ogrp@scm01:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         172.30.0.1      0.0.0.0         UG    512    0        0 eth0
10.3.0.0        0.0.0.0         255.255.255.0   U     0      0        0 eth1
10.3.1.0        0.0.0.0         255.255.255.0   U     0      0        0 eth2
10.3.2.0        0.0.0.0         255.255.255.0   U     0      0        0 eth3
10.3.3.0        0.0.0.0         255.255.255.0   U     0      0        0 eth4
10.3.4.0        0.0.0.0         255.255.255.0   U     0      0        0 eth5
10.3.5.0        0.0.0.0         255.255.255.0   U     0      0        0 eth6
10.3.6.0        0.0.0.0         255.255.255.0   U     0      0        0 eth7
10.3.7.0        0.0.0.0         255.255.255.0   U     0      0        0 eth8
10.3.8.0        0.0.0.0         255.255.255.0   U     0      0        0 eth9
10.3.9.0        0.0.0.0         255.255.255.0   U     0      0        0 eth10
10.3.10.0       0.0.0.0         255.255.255.0   U     0      0        0 eth11
10.4.0.0        0.0.0.0         255.255.255.0   U     0      0        0 eth12
10.4.1.0        0.0.0.0         255.255.255.0   U     0      0        0 eth13
10.4.2.0        0.0.0.0         255.255.255.0   U     0      0        0 eth14
172.30.0.0      0.0.0.0         255.255.0.0     U     0      0        0 eth0
172.30.0.1      0.0.0.0         255.255.255.255 UH    512    0        0 eth0
192.168.254.0   172.30.0.1      255.255.255.0   UG    99     0        0 eth0
csci3306ogrp@scm01:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.30.0.21  netmask 255.255.0.0  broadcast 172.30.255.255
        inet6 fe80::f0aa:e9ff:fe6f:afa4  prefixlen 64  scopeid 0x20<link>
        ether f2:aa:e9:6f:af:a4  txqueuelen 1000  (Ethernet)
        RX packets 1044  bytes 538474 (538.4 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 431  bytes 51835 (51.8 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.3.0.1  netmask 255.255.255.0  broadcast 10.3.0.255
        inet6 fe80::1441:32ff:fe0e:6999  prefixlen 64  scopeid 0x20<link>
        ether 16:41:32:0e:69:99  txqueuelen 1000  (Ethernet)
        RX packets 49  bytes 3882 (3.8 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 15  bytes 1146 (1.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.3.1.1  netmask 255.255.255.0  broadcast 10.3.1.255
        inet6 fe80::d48a:c6ff:fef7:accb  prefixlen 64  scopeid 0x20<link>
        ether d6:8a:c6:f7:ac:cb  txqueuelen 1000  (Ethernet)
        RX packets 50  bytes 3992 (3.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 15  bytes 1146 (1.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.3.2.1  netmask 255.255.255.0  broadcast 10.3.2.255
        inet6 fe80::a8ac:56ff:fe89:376d  prefixlen 64  scopeid 0x20<link>
        ether aa:ac:56:89:37:6d  txqueuelen 1000  (Ethernet)
        RX packets 49  bytes 3922 (3.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 15  bytes 1146 (1.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth4: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
```

Most of the nodes looked like the one on the left. Each node has multiple interfaces and each one is associated with different IPs and subnet masks, and they also include a loopback. The default is set to go through “eth0”. The “scm01” node has more interfaces, up to “eth14”, but it still shares the default.

Redrawing the topology



I redrew the topology with some IPs. It's very hard to tell what every single interface is between each node, but I provided at least 1 for all nodes. Note "scm01" is the most complete since it faces the "landsat" nodes on one side.

Adding connectivity between unconnected nodes

```
csci3306ogrp@soc:~$ sudo route add -net 10.5.2.0 netmask 255.255.255.0 gw 10.5.2.1 dev eth1
SIOCADDRT: File exists
csci3306ogrp@soc:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         172.30.0.1      0.0.0.0         UG    512    0        0 eth0
10.5.2.0        10.5.2.1        255.255.255.0   UG    0      0        0 eth1
10.5.2.0        0.0.0.0         255.255.255.0   U     0      0        0 eth1
172.30.0.0      0.0.0.0         255.255.0.0     U     0      0        0 eth0
172.30.0.1      0.0.0.0         255.255.255.255 UH    512    0        0 eth0
192.168.254.0   172.30.0.1      255.255.255.0   UG    99     0        0 eth0
csci3306ogrp@soc:~$
```

I added connectivity between "soc" and "socgw", which were previously not connected. To do this I used "sudo route add -net 10.5.2.0 netmask 255.255.255.0 gw 10.5.2.1 dev eth1". Notice the change is now reflected on the "soc" route.

"dublin" to "newalbany", "seoul" to "landsat9" (failed):

```
csci3306ogrp@dublin:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         172.30.0.1      0.0.0.0         UG    512    0        0 eth0
10.1.7.0        0.0.0.0         255.255.255.0   U     0      0        0 eth1
10.5.1.0        0.0.0.0         255.255.255.0   U     0      0        0 eth2
172.30.0.0      0.0.0.0         255.255.0.0     U     0      0        0 eth0
172.30.0.1      0.0.0.0         255.255.255.255 UH    512    0        0 eth0
192.168.254.0   172.30.0.1      255.255.255.0   UG    99     0        0 eth0
csci3306ogrp@dublin:~$ ping 10.1.9.1
PING 10.1.9.1 (10.1.9.1) 56(84) bytes of data.
^C
--- 10.1.9.1 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2038ms
```

```
csci3306ogrp@dublin:~$ sudo ip route add 10.1.9.0 dev eth1
csci3306ogrp@dublin:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         172.30.0.1      0.0.0.0         UG    512    0        0 eth0
10.1.7.0        0.0.0.0         255.255.255.0   U     0      0        0 eth1
10.1.9.0        0.0.0.0         255.255.255.255 UH    0      0        0 eth1
10.5.1.0        0.0.0.0         255.255.255.0   U     0      0        0 eth2
172.30.0.0      0.0.0.0         255.255.0.0     U     0      0        0 eth0
172.30.0.1      0.0.0.0         255.255.255.255 UH    512    0        0 eth0
192.168.254.0   172.30.0.1      255.255.255.0   UG    99     0        0 eth0
csci3306ogrp@dublin:~$
```

```
csci3306ogrp@dublin:~$ sudo route add -net 10.1.9.0 netmask 255.255.255.0 gw 10.1.9.1 dev eth1
SIOCADDRT: Network is unreachable
```

```
csci3306ogrp@seoul:~$ sudo ip route add 10.2.1.0 dev eth1
csci3306ogrp@seoul:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         172.30.0.1      0.0.0.0         UG    512    0        0 eth0
10.1.0.0        0.0.0.0         255.255.255.0   U     0      0        0 eth1
10.2.1.0        0.0.0.0         255.255.255.255 UH    0      0        0 eth1
10.5.1.0        0.0.0.0         255.255.255.0   U     0      0        0 eth2
172.30.0.0      0.0.0.0         255.255.0.0     U     0      0        0 eth0
172.30.0.1      0.0.0.0         255.255.255.255 UH    512    0        0 eth0
192.168.254.0   172.30.0.1      255.255.255.0   UG    99     0        0 eth0
csci3306ogrp@seoul:~$ sudo route add -net 10.2.1.0 netmask 255.255.255.0 gw 10.2.1.1 dev eth1
SIOCADDRT: Network is unreachable
csci3306ogrp@seoul:~$
```

To begin, I tried pinging one of "newalbany"'s IPs. Notice destination "10.1.9.0" is not on "dublin"'s route. I added it using the command above. (I later realized this was because the "10.1.9.0" is meant for "scm01" and all its interfaces were "10.*.1.0" connecting it to all the city named nodes). I noticed the "landsat*" nodes were only connected to "scm01" so I thought maybe I could connect one of those nodes to a city named node, but after adding "10.2.1.0" into the routing table, I was also unsuccessful. I tried with other city named nodes as well and they were actually able to ping each other. I will provide some proof of this below. After further inspection, I noticed the changes between the provided topology and the topology described in the paper were significant. In the provided topology, the city named nodes were all connected by LAN.

“singapore” pings “lima”:

```
csci3306ogrp@singapore:~$ ping 10.5.1.10
PING 10.5.1.10 (10.5.1.10) 56(84) bytes of data.
64 bytes from 10.5.1.10: icmp_seq=1 ttl=64 time=1.01 ms
64 bytes from 10.5.1.10: icmp_seq=2 ttl=64 time=0.413 ms
64 bytes from 10.5.1.10: icmp_seq=3 ttl=64 time=0.445 ms
^C
--- 10.5.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2021ms
rtt min/avg/max/mdev = 0.413/0.623/1.011/0.274 ms
csci3306ogrp@singapore:~$
```

“lima” pings “stockholm”:

```
csci3306ogrp@lima:~$ ping 10.5.1.4
PING 10.5.1.4 (10.5.1.4) 56(84) bytes of data.
64 bytes from 10.5.1.4: icmp_seq=1 ttl=64 time=0.932 ms
64 bytes from 10.5.1.4: icmp_seq=2 ttl=64 time=0.360 ms
64 bytes from 10.5.1.4: icmp_seq=3 ttl=64 time=0.391 ms
^C
--- 10.5.1.4 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2022ms
rtt min/avg/max/mdev = 0.360/0.561/0.932/0.262 ms
csci3306ogrp@lima:~$
```

“stockholm” pings “capetown”:

```
csci3306ogrp@stockholm:~$ ping 10.5.1.8
PING 10.5.1.8 (10.5.1.8) 56(84) bytes of data.
64 bytes from 10.5.1.8: icmp_seq=1 ttl=64 time=0.900 ms
64 bytes from 10.5.1.8: icmp_seq=2 ttl=64 time=0.408 ms
64 bytes from 10.5.1.8: icmp_seq=3 ttl=64 time=0.366 ms
^C
--- 10.5.1.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2014ms
rtt min/avg/max/mdev = 0.366/0.558/0.900/0.242 ms
csci3306ogrp@stockholm:~$
```

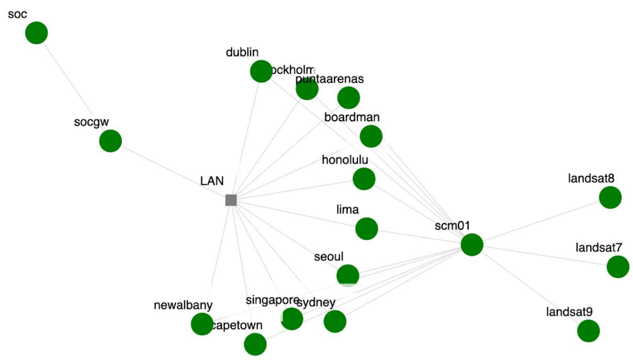
“capetown” pings “puntaarenas”:

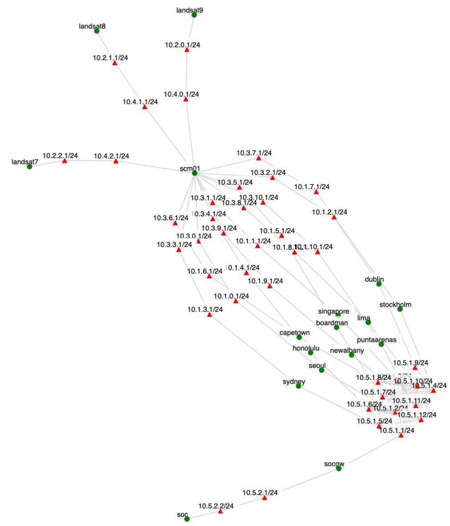
```
csci3306ogrp@capetown:~$ ping 10.5.1.12
PING 10.5.1.12 (10.5.1.12) 56(84) bytes of data.
64 bytes from 10.5.1.12: icmp_seq=1 ttl=64 time=0.991 ms
64 bytes from 10.5.1.12: icmp_seq=2 ttl=64 time=0.388 ms
64 bytes from 10.5.1.12: icmp_seq=3 ttl=64 time=0.392 ms
^C
--- 10.5.1.12 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2017ms
rtt min/avg/max/mdev = 0.388/0.590/0.991/0.283 ms
csci3306ogrp@capetown:~$
```

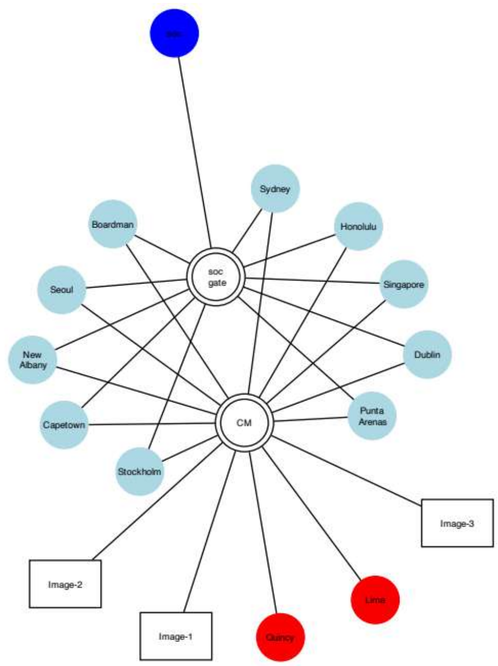
“puntaarenas” pings “honolulu”:

```
csci3306ogrp@puntaarenas:~$ ping 10.5.1.6
PING 10.5.1.6 (10.5.1.6) 56(84) bytes of data.
64 bytes from 10.5.1.6: icmp_seq=1 ttl=64 time=0.355 ms
64 bytes from 10.5.1.6: icmp_seq=2 ttl=64 time=0.393 ms
64 bytes from 10.5.1.6: icmp_seq=3 ttl=64 time=0.395 ms
^C
--- 10.5.1.6 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2049ms
rtt min/avg/max/mdev = 0.355/0.381/0.395/0.018 ms
csci3306ogrp@puntaarenas:~$
```

Topology Difference







Notice the LAN connection I mentioned between all city named nodes. In the second picture, this is further represented since all the nodes are connected to “socgw” through “10.5.1.1”. In the last image, there is a visible difference in the connections of the topology.

Summary

For this Lab, I used a provided topology on an XDC to simulate a network and access all its nodes. I had no problem accessing everything and installing the required “net-tools”. The difference between managing the connections through Linux and GNS3 was easily noticed. On GNS3, you have to configure each router and interface individually. The process is further complicated sometimes because of the number of IP addresses that we must set to connect all routers accordingly. However, I still feel more comfortable using GNS3 commands, likely due to how many times I’ve had to repeat them for past labs. Going back to this lab, I’m not sure I accomplished what was intended since as it appears to me, the nodes were already connected. I still thought it was interesting that “soc” and “socgw” worked. If I were more experienced in networking, I might’ve been able to successfully connect all nodes in some other way. However, my inexperience matched with my unfamiliarity with the Linux “net-tools” commands ultimately led me to fail.